Final Project Report

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Overview

For my project, I tested three reinforcement learning (RL) algorithms (DQN, PPO and A2C) in three different environments (highway-v0: easy; roundabout-v0: medium; and intersection-v0: hard), and did extensive hyperparameter tuning to find the best possible settings for the driving task. Stable-Baselines3 was used as the implemented RL framework. First, baseline models were created for each of the three environments with the default parameter values. Then, I looked at the charts and videos of the baseline to see what changes could be made to the hyperparameters to increase the agent's performance.

Initial Hypothesis

I expect PPO to achieve the highest rewards in the intersection and roundabout environments because its clipped surrogate objective prevents policy collapse, where the agent suddenly becomes random or unstable. A2C will likely train the fastest due to its synchronous updates but may show higher variance. DQN is expected to perform well on the straight highway but might struggle with the complex coordination required for the roundabout due to its reliance on a Q-table/Value-function approximation in highly dynamic states.

Environments and Algorithms

Highway-env is a Gymnasium-based suite of driving environments. Three environments were selected to span a difficulty range:

- highway-v0: A straight multi-lane highway. The agent must navigate traffic by changing lanes and keeping speed. The highway is the easiest environment because of its predictable, unidirectional structure.
- roundabout-v0: A circular roundabout intersection. The agent must time entry into the roundabout, navigate the curve, and exit safely. Therefore, the agent requires more spatial awareness than the highway.
- intersection-v0: four-way intersection with cross-traffic. The agent must yield to perpendicular vehicles. The intersection is the hardest environment because of sparse rewards and the need for patient, timing-based decisions.

Three algorithms from Stable-Baselines3 were used: DQN (Deep Q-Network), a value-based method that uses experience replay and a target network; PPO (Proximal Policy Optimization), a policy-gradient method that uses a clipped surrogate objective to limit the size of policy updates;

and A2C (Advantage Actor-Critic), an actor-critic method that uses synchronous updates and frequent gradient steps.

Methodology

Each training run was 25,000 timesteps. Models were saved as .zip checkpoints to Google Drive after every 5,000 steps to allow protection against Colab session disconnections, which were extremely common during this project. After training, evaluation was performed over one complete episode with the agent operating deterministically. Videos were recorded for each evaluated model using imageio. Metrics (total reward and survival steps) were stored as JSON files.

The overall pipeline for each environment was: (1) Train all three algorithms with default hyperparameters; (2) Evaluate and record baseline videos and metrics; (3) Examine the charts and video behaviors to determine the primary failure modes; (4) Set tuned hyperparameters based upon the observations; (5) Retrain with the new hyperparameters; (6) Evaluate and record the tuned videos and metrics; and (7) Compare baseline vs. tuned results.

The total run-time of all 18 training runs (three environments x three algorithms x baseline and tuned) was over thirty hours. The total run-time for the highway environment was about nine hours for all three algorithms. The total run-time for both the roundabout and intersection environments was about three hours for each algorithm.

Highway-v0 (Easy)

Baseline Training and Observations

All three algorithms were trained with default Stable-Baselines3 hyperparameters. The training All three algorithms were trained using the default hyperparameters of Stable-Baselines3. The training charts (Fig. 1) display the mean of the episode reward and mean of the episode length over the course of training. The key observation from the charts is that:

- PPO showed steady reward growth from early in training, stabilizing near its maximum well before 25,000 steps.
- A2C showed highly erratic reward curves with frequent large drops, consistent with the high-variance nature of on-policy updates without a clipping mechanism.
- DQN showed a slow initial learning period (typical of the epsilon-greedy warmup phase) before gradually improving in the second half of training.

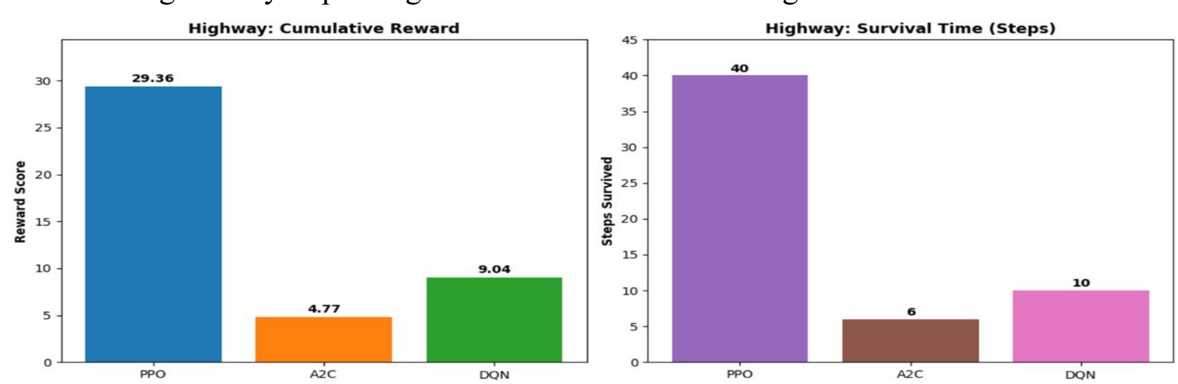
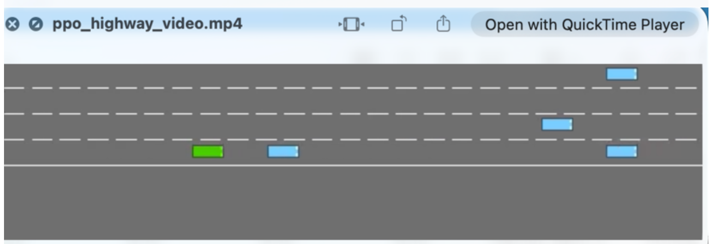


Figure 1: Highway-v0 Baseline Training - Rewards & Survival Steps (PPO, A2C, DQN)

When evaluating each of the baseline models, a common failure mode was observed among all three algorithms. Each agent drove down the middle of its lane the whole episode without changing lanes. In addition, each agent eventually collided with the slower traffic ahead of it without making any attempt to evade.

This represents a classic case of insufficient exploration. Under the default hyperparameters, neither of the algorithms explored a sufficient portion of the state-action space to recognize that switching lanes was a viable strategy for avoiding slower traffic. Instead, each agent learned a locally stable policy (stay in lane, maintain speed) that yielded moderate reward via survival duration but failed to learn the globally better policy of proactively managing their lane.



Environment / Metric	PPO	A2C	DQN
Highway-v0 – Reward	29.36	4.77	9.04
Highway-v0 – Survival Steps	40	6	10

Table 1: Highway-v0 Baseline Evaluation Results

Hyperparameter Tuning Decisions

Based on the baseline observations, the central goal of tuning was to encourage all three algorithms to explore lane-changing behavior. The specific tuning decisions per algorithm were:

DQN – Highway Tuning

Parameter	Default	Tuned	Change
learning_rate	0.0001	0.0005	+5x
exploration_fraction	0.1	0.5	+5x
batch_size	128	256	2x

gamma	0.99	0.8	Lowered
buffer_size	1,000,000	100,000	Reduced

Table 2: DQN Tuned Hyperparameters - Highway-v0

The biggest change was the exploration fraction. By increasing this fraction from 0.1 to 0.5, DQN gets 50% more of its 25,000 step training period to spend exploring with random actions and observing how they affect the outcome. The learning rate was also increased from 0.0001 to 0.0005 to help speed up learning. Also the batch size was increased to 256 for better estimates of the gradients. As the highway is a forward moving structure, gamma was decreased to 0.8 so that DQN will focus on immediate rewards for going faster. Lastly the buffer size was decreased from the default 1,000,000 to 100,000. The reason for decreasing the buffer size was to keep the replay buffer full of recent and relevant data for the short training run.

PPO – Highway Tuning

Parameter	Default	Tuned	Change
learning_rate	0.0003	0.0003	Unchanged
n_steps	2048	1024	Halved
batch_size	64	128	2x
gamma	0.99	0.99	Unchanged
ent_coef	0.0	0.01	Added

Table 3: PPO Tuned Hyperparameters - Highway-v0

Since the baseline already had good results (reward = 29.36, steps = 40), the main objective for PPO's tuning was to see if there was room for improvement. The best modification for PPO was to add a small entropy coefficient (0.01) to discourage PPO from becoming too deterministic as soon as possible. Decreasing n_steps from 2048 to 1024 allows the policy to update more frequently, which is useful for environments like highways, where the environment can change rapidly. The learning rate and gamma were kept at their standard PPO values.

A2C – Highway Tuning

Parameter	Default	Tuned	Change
learning_rate	0.0007	0.0005	Slightly reduced
n_steps	5	10	2x
gamma	0.99	0.99	Unchanged
ent_coef	0.0	0.01	Added
rms_prop_eps	1e-5	1e-5	Unchanged

Table 4: A2C Tuned Hyperparameters - Highway-v0

As A2C's baseline was very bad (reward = 4.77, 6 steps), A2C failed completely to learn lane-keeping. Increasing n_steps from 5 to 10 gives a slightly larger batch of experience per update than before, which reduces some of the noise that was producing the wild reward curves of A2C.

A small entropy coefficient (0.01) was added to PPO to prevent premature convergence. The learning rate was slightly decreased from 0.0007 to 0.0005 to improve stability of the very noisy gradients.

Tuned Results and Analysis

Environment / Metric	PPO	A2C	DQN
Highway-v0 Baseline – Reward	29.36	4.77	9.04
Highway-v0 Tuned – Reward	29.36	29.36	11.69
Highway-v0 Baseline – Steps	40	6	10
Highway-v0 Tuned – Steps	40	40	13

Table 5: Highway-v0 Baseline vs. Tuned Evaluation Results

The largest change was A2C, which went from a baseline reward of 4.77 (6 survival steps) to a reward of 29.36 (40 steps) for the tuned version. This shows that A2C's poor performance was not an issue with the A2C algorithm, but rather the hyperparameters used to train A2C. When A2C was trained with the large enough n_steps buffer and a stable learning rate, A2C achieved the same performance as PPO.

DQN was able to improve from 9.04 to 11.69 (an increase of 29%, or 3 additional survival steps) from the baseline. Most importantly, the video of the tuned DQN showed qualitative differences from the videos of all the other agents. DQN was the first agent to demonstrate active lane switching. DQN started in lane 2, switched lanes to 3 when a vehicle appeared ahead of it, then switched to lane 4 and continued navigating. Eventually it crashed after a couple of lane switches, but the fact that DQN demonstrated proactive lane management is exactly what the tuning was intended to produce.

PPO’s tuned performance was identical to its baseline (29.36, 40 steps). This indicates that PPO’s default configuration was already optimal for the highway environment, and the small entropy and n_steps changes did not provide additional benefit beyond what the baseline had already achieved.

Roundabout-v0 (Medium)

Baseline Training and Observations

Unlike the highway, where baseline performance varied widely across algorithms, all three algorithms clustered in a narrow reward band at baseline (9.17–10.08). This indicates that the roundabout has a relatively accessible reward floor that all three algorithms can reach within 25,000 steps, but also that none of them are fully solving the environment.

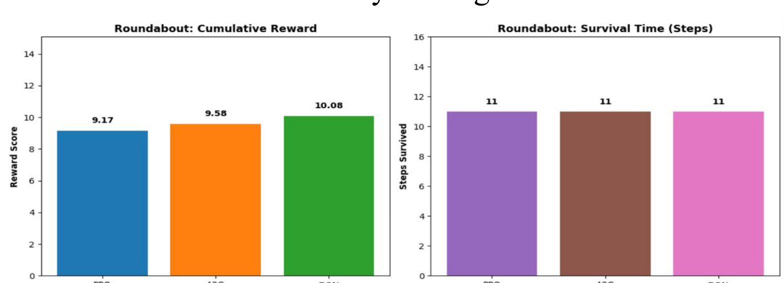
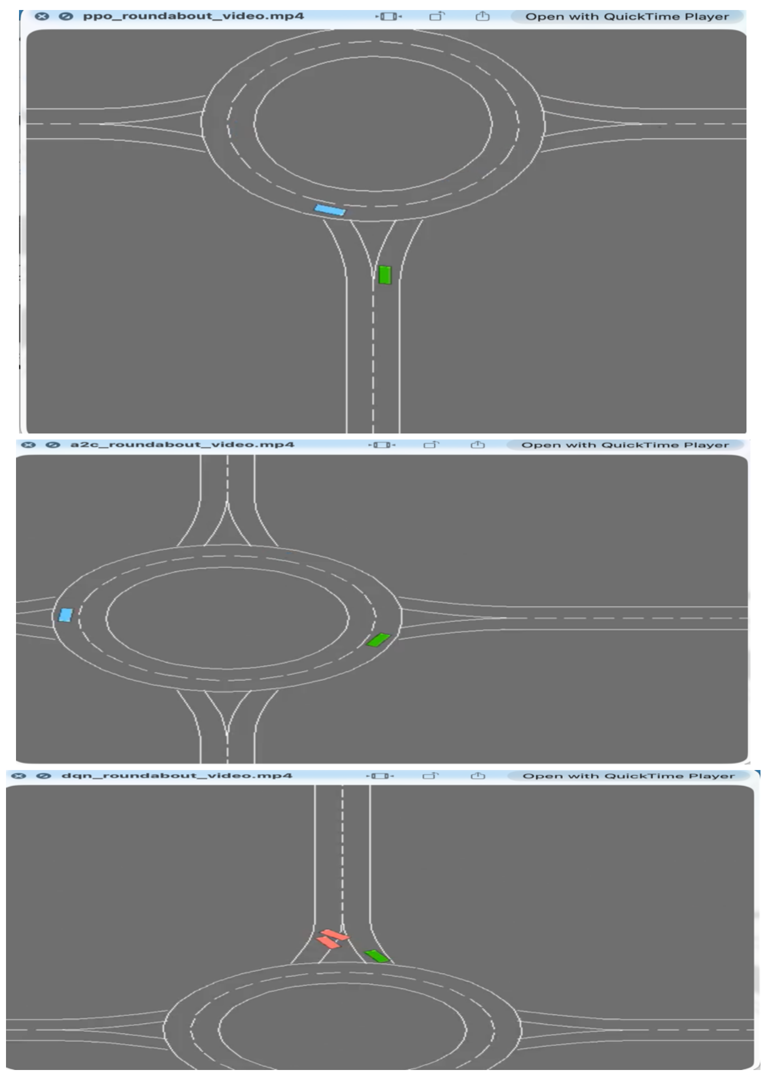


Figure 2: Roundabout-v0 Baseline Training – Rewards & Survival Steps (PPO, A2C, DQN)

The survival steps were identical across all three algorithms (11 steps each), which is an important observation: all agents are surviving for the same duration, so the reward differences arise not from how long they survive but from how efficiently they navigate the roundabout during those steps. An agent that moves smoothly and quickly accumulates more reward per step than one that hesitates or moves inefficiently. The video observations revealed starkly different behaviors across algorithms:

- PPO: Barely moved from its starting position at the roundabout entrance. It remained almost stationary for most of the episode, receiving minimal reward from the sparse forward-movement signal.
- A2C: Entered from the bottom-right, traveled approximately one quarter of the roundabout arc, then stopped. It did not complete even half of the circuit.
- DQN: Entered from the bottom-right, successfully navigated all the way through the roundabout to the opposite straight road, and exited cleanly. Two other vehicles in the environment collided with each other, but the DQN agent's vehicle was not involved. DQN's higher baseline reward (10.08) is directly explained by this more complete navigation.



Environment / Metric	PPO	A2C	DQN
Roundabout-v0 – Reward	9.17	9.58	10.08
Roundabout-v0 – Survival Steps	11	11	11

Table 6: Roundabout-v0 Baseline Evaluation Results

Hyperparameter Tuning Decisions

The baseline videos revealed that the central failure across all three algorithms was insufficient exploration of the yielding and merging behaviors required by the roundabout. The tuning strategy was to increase the entropy coefficient for PPO and A2C (forcing more exploratory policies) and to increase gamma for all algorithms (to make agents value the future reward of completing the circuit rather than settling for minimal forward movement).

DQN – Roundabout Tuning

Parameter	Default	Tuned	Rationale
learning_rate	0.0001	0.0005	Faster learning within budget
exploration_fraction	0.1	0.5	More time to discover merge/exit behaviors
batch_size	128	256	Stable gradient estimates
gamma	0.99	0.9	Slightly lower: balance near-term merge timing vs. future
buffer_size	1,000,000	100,000	Recent experience more relevant in dynamic roundabout

Table 7: DQN Tuned Hyperparameters - Roundabout-v0

Gamma was set to 0.9 for the roundabout (compared to 0.8 for the highway) because the roundabout requires the agent to look slightly further ahead to successfully time the merge and plan the exit. A higher exploration fraction (0.5) was again the most critical change, giving DQN significantly more time to discover the state-action combinations associated with roundabout entry and circuit navigation.

PPO – Roundabout Tuning

Parameter	Default	Tuned	Rationale
learning_rate	0.0003	0.0003	Standard PPO rate, kept unchanged
n_steps	2048	1024	More frequent updates for dynamic environment
batch_size	64	128	Larger mini-batch for stable updates
gamma	0.99	0.95	Near-term merge timing is more important than far future
ent_coef	0.0	0.05	High entropy: force exploration of yield-and-wait behaviors

Table 8: PPO Tuned Hyperparameters - Roundabout-v0

The most significant change for PPO was the entropy coefficient, increased from 0.0 to 0.05. The baseline video showed PPO was almost completely stationary a sign that it had converged to a near-zero movement policy very early in training. High entropy (0.05) forces PPO to keep exploring rather than committing to this degenerate policy. Gamma was lowered to 0.95 because the roundabout requires prioritizing the near-term reward of merge timing over a distant future reward.

A2C – Roundabout Tuning

Parameter	Default	Tuned	Rationale
learning_rate	0.0007	0.0005	Reduced to stabilize noisy updates
n_steps	5	10	Slightly larger buffer reduces update noise
gamma	0.99	0.95	Near-term merge timing priority
ent_coef	0.0	0.05	Force exploration of yielding behaviors
rms_prop_eps	1e-5	1e-5	Precise gradient scaling for noisy reward signal

Table 9: A2C Tuned Hyperparameters - Roundabout-v0

A2C received the same entropy and gamma adjustments as PPO for the roundabout, reflecting the same underlying problem: insufficient exploration of yield-and-wait behaviors. The rms_prop_eps value of 1e-5 was kept from the highway tuning as it provides stable gradient scaling in the RMSProp optimizer under the noisy reward signals characteristic of the roundabout.

Tuned Results and Analysis

Environment / Metric	PPO	A2C	DQN
Roundabout-v0 Baseline -Reward	9.17	9.58	10.08
Roundabout-v0 Tuned - Reward	9.17	11.0	10.58
Roundabout-v0 Baseline - Steps	11	11	11
Roundabout-v0 Tuned –-Steps	11	11	11

Table 10: Roundabout-v0 Baseline vs. Tuned Evaluation Results

A2C showed the best tuned performance, improving from 9.58 to 11.0. The high entropy coefficient (0.05) successfully pushed A2C to explore yield-and-wait behaviors it had not discovered at baseline. The tuned video confirms this: A2C entered from the bottom road, navigated the full roundabout circuit, reached the opposite straight road, and continued past the exit. This is a meaningfully more complete navigation than the baseline.

DQN improved from 10.08 to 10.58, a modest but real improvement. The tuned DQN video showed the same behavior as baseline (entering, navigating the arc, exiting to the straight road) but with slightly smoother execution, reflected in the higher reward per step.

PPO's tuned performance was identical to its baseline (9.17, 11 steps), despite the significant entropy coefficient change. This is a striking result. One interpretation is that PPO's near-stationary behavior at baseline was already its converged policy for this environment given the training budget, and that 25,000 steps is not sufficient for the entropy-driven exploration to discover and reinforce the yield-and-merge strategy. This represents a training budget limitation rather than an algorithmic one.

Intersection-v0 (Hard)

Baseline Training and Observations

The intersection baseline results were the most variable of the three environments. A2C produced a surprisingly high positive reward (9.0) while PPO scored 0.0 and DQN scored -1.0, making A2C appear to be the best algorithm for this environment.

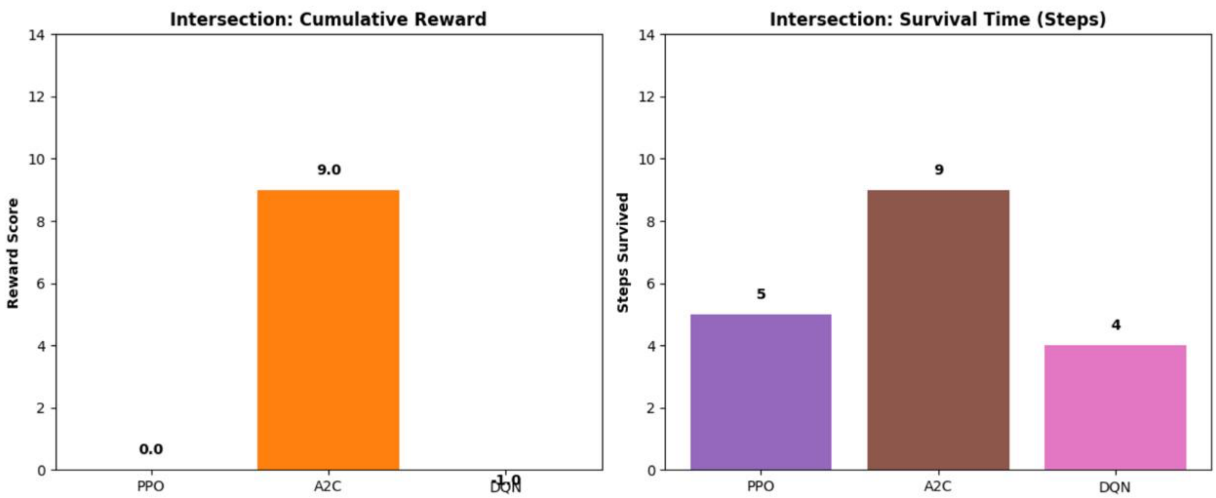


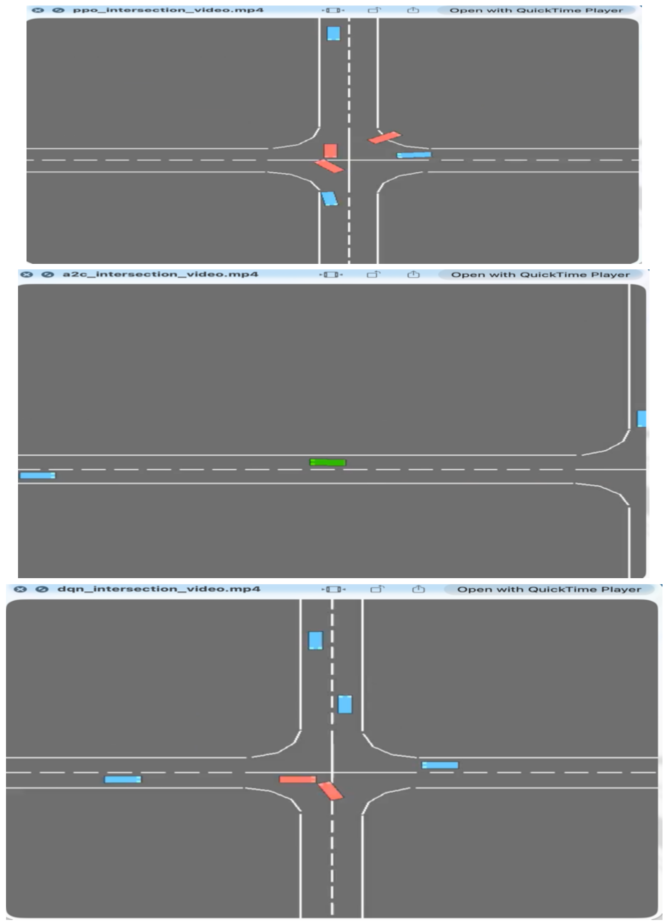
Figure 3: Intersection-v0 Baseline Training – Rewards & Survival Steps (PPO, A2C, DQN)

However, the video observations tell a more nuanced story. All three baseline agents approached from the bottom-right lane and attempted an immediate left turn without any waiting or yielding behavior:

- PPO: Attempted the left turn and collided with a vehicle already involved in a prior crash. The blocking vehicles were stationary wrecks in the path of the turn; the collision was unavoidable given the approach trajectory.
- A2C: Turned left and passed through the intersection without any collision. Critically, this occurred because there were no vehicles present in the intersection at the moment the agent passed through this was a lucky episode, not a learned strategy.
- DQN: Attempted the left turn and collided with an oncoming vehicle coming from the side.

The key insight from these videos is that A2C's 9.0 baseline reward was entirely situational a fortuitous absence of cross-traffic, not a learned waiting behavior. None of the three algorithms showed any evidence of looking, waiting, or timing their crossing. This is consistent with the exploration-exploitation challenge: the correct strategy (stop and wait for a gap) yields zero reward in the short term, while any forward movement generates at least a small reward signal.

Without explicit incentive or enough exploration to accidentally discover the waiting strategy, all agents default to immediate forward movement.



Environment / Metric	PPO	A2C	DQN
Intersection-v0 – Reward	0.0	9.0	-1.0
Intersection-v0 – Survival Steps	5	9	4

Table 11: Intersection-v0 Baseline Evaluation Results

Hyperparameter Tuning Decisions

The core challenge identified in the baseline was the sparse reward problem: the agent needs to value the future benefit of waiting (avoiding a crash that may happen 5–10 steps later) over the small immediate reward of moving forward. The tuning strategy was to maximize gamma (to make agents care deeply about long-term outcomes) and increase entropy (to force exploration of non-default actions such as stopping or slowing down).

DQN – Intersection Tuning

Parameter	Default	Tuned	Rationale
learning_rate	0.0001	0.0005	Faster learning within budget
exploration_fraction	0.1	0.5	More exploration of braking/waiting behaviors
batch_size	128	256	Stable gradient estimates
gamma	0.99	0.95	High: value future survival over immediate progress
buffer_size	1,000,000	100,000	Relevant recent experience only

Table 12: DQN Tuned Hyperparameters – Intersection-v0

Gamma was set to 0.95 for the intersection, higher than for the roundabout (0.9), because the intersection requires the agent to wait for events that may be several steps away. A discount factor of 0.95 means a reward 20 steps in the future is still worth 0.95^20 =0.36 of its full value enough to motivate waiting behavior if the agent encounters a successful crossing episode during training. The extended exploration fraction was again the key change to give DQN sufficient time to discover waiting behaviors.

PPO and A2C – Intersection Tuning

Parameter	PPO Tuned	A2C Tuned	Rationale
learning_rate	0.0003	0.0005	Standard rates for each algorithm
n_steps	1024	10	PPO: large rollout; A2C: frequent updates
batch_size (PPO)	128	-	Standard PPO mini-batch
gamma	0.99	0.99	Maximum: force the agent to look far into the future
ent_coef	0.05	0.05	High entropy: explore waiting/yielding
rms_prop_eps (A2C)	-	1e-5	Precise gradient scaling

Table 13: PPO and A2C Tuned Hyperparameters – Intersection-v0

For both PPO and A2C, gamma was set to 0.99 the maximum to force the agents to treat future survival as nearly as valuable as immediate reward. This is theoretically the correct choice for the intersection: an agent must look far enough into the future to see that crashing in 8 steps is a very bad outcome worth avoiding right now. The entropy coefficient of 0.05 was maintained from the roundabout tuning to continue pushing exploration of non-obvious behaviors.

Tuned Results and Analysis

Environment / Metric	PPO	A2C	DQN
Intersection-v0 Baseline – Reward	0.0	9.0	-1.0
Intersection-v0 Tuned – Reward	0.0	0.0	0.0
Intersection-v0 Baseline – Steps	5	9	4

Intersection-v0 Tuned – Steps	5	6	6

Table 14: Intersection-v0 Baseline vs. Tuned Evaluation Results

After tuning, all three algorithms converged to 0.0 reward and 5–6 survival steps. The tuned videos confirmed the same behavior as baseline: all three agents drove directly into the intersection without any waiting or yielding, and all three collided.

- PPO (tuned): Collided with an oncoming vehicle entering the intersection from the left side road.
- A2C (tuned): Collided with already-crashed vehicles blocking the turning path the same scenario as the baseline PPO.
- DQN (tuned): Collided with a vehicle coming straight from above at the intersection center.

The convergence to 0.0 represents a slight improvement for PPO (from 0.0 to 0.0, unchanged) and DQN (from -1.0 to 0.0) but a decline for A2C (from 9.0 to 0.0). The decline in A2C's performance is actually informative: the 9.0 baseline result was a lucky episode that had no vehicles in the intersection, while the 0.0 tuned result reflects a more typical episode where cross-traffic is present. The tuned metrics are a more honest representation of the algorithm's actual capability.

The fundamental issue is a sparse reward problem compounded by a limited training budget. The correct strategy for the intersection requires the agent to (1) approach and slow down, (2) observe the gap in cross-traffic, (3) wait 5–10 steps with zero reward accumulation, and (4) proceed through the gap. This is a temporally extended behavior that epsilon-greedy or entropy-driven exploration is very unlikely to accidentally produce and complete within 25,000 timesteps. Even if the agent stumbles into the waiting behavior, it needs to encounter enough successful crossing episodes to reinforce the correct sequence which 25,000 steps is insufficient to guarantee.

This directly illustrates the exploration-exploitation dilemma discussed in class. In environments where the reward signal for the correct action (waiting) is zero or even slightly negative in the short term, and where the penalty for the incorrect action (crossing at the wrong time) is a large negative terminal reward, agents will systematically underexplore waiting behaviors. Reward shaping for example, giving small positive rewards for slowing down, maintaining safe following distance, or identifying gaps in cross-traffic would likely be necessary to solve this environment within the given budget.

Overall Comparison: Baseline vs. Tuned

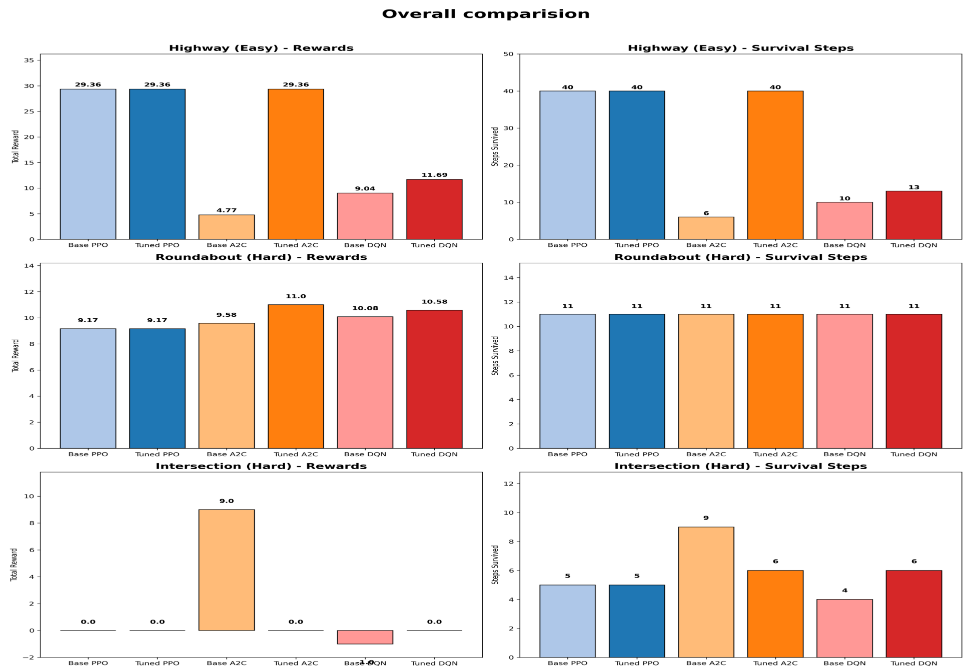


Figure 4

Algorithm	Highway Base	Highway Tuned	Roundabout Base	Roundabout Tuned	Intersection Base	Intersection Tuned
PPO Reward	29.36	29.36	9.17	9.17	0.0	0.0
A2C Reward	4.77	29.36	9.58	11.0	9.0*	0.0
DQN Reward	9.04	11.69	10.08	10.58	-1.0	0.0

Table 15: Complete Results Summary (All Environments, Baseline vs. Tuned) - A2C intersection baseline was situational (lucky clear episode)

PPO was the most consistently stable performer across all environments and difficulty levels. It achieved the maximum highway reward at baseline, never collapsed or degraded after tuning, and maintained performance across all three environments. This confirms the theoretical value of PPO's clipped surrogate objective: by limiting the magnitude of each policy update, PPO avoids the catastrophic policy collapses that afflicted A2C at baseline and the slow convergence that characterized DQN.

A2C was the most tuning-sensitive algorithm. It produced the worst highway baseline (4.77) but, after tuning, matched PPO's maximum highway performance (29.36) a 515% improvement. It also produced the best roundabout tuned performance (11.0). This sensitivity is consistent with A2C's on-policy, high-frequency update mechanism: small changes to learning rate, n_steps, or entropy can dramatically change whether the algorithm converges to a good or bad policy. A2C's high variance is both its weakness (unpredictable baseline performance) and its strength (large potential for improvement through targeted tuning).

DQN provided reliable but moderate improvements across all environments. Its tuning improvements were the smallest of the three algorithms (29% on highway, 5% on roundabout, -1.0 to 0.0 on intersection). This is consistent with value-based methods having a lower adaptive ceiling in dynamic, multi-agent environments. DQN's reliance on a discrete Q-table approximation becomes less accurate as the environment state space grows more complex, and its epsilon-greedy exploration strategy requires significant training time to discover the nuanced behavioral patterns needed in the roundabout and intersection.

The difficulty gradient across environments was clear and confirmed the hypothesis. The highway was fully learnable by all algorithms within 25,000 timesteps, with all three reaching at least partial performance at baseline and all improving after tuning. The roundabout was partially learnable all algorithms reached a reward band of 9–11, but no algorithm fully solved the environment. The intersection was largely unsolvable within the training budget, with all algorithms converging to 0.0 reward in the tuned evaluation regardless of hyperparameter configuration.

Revisiting the Initial Hypothesis

The initial hypothesis that PPO would achieve the highest rewards in the intersection and roundabout was partially supported. PPO was the dominant baseline performer on the highway (29.36 vs 4.77 for A2C and 9.04 for DQN) and maintained its stability across all environments, which confirms that the clipped surrogate objective effectively prevents policy collapse. However, in the roundabout, DQN edged out PPO at baseline (10.08 vs 9.17), and A2C outperformed both at tuning (11.0). In the intersection, A2C produced the only positive baseline reward (9.0), which was the opposite of what was predicted for PPO.

The prediction that A2C would train fastest but show higher variance was confirmed. A2C produced the largest performance swing of any algorithm: worst on highway at baseline (4.77), best on intersection at baseline (9.0), and best on roundabout after tuning (11.0). The frequent synchronous updates made it highly sensitive to the right hyperparameter configuration, and the variance in results across environments is exactly what was predicted.

The prediction about DQN was partially confirmed. DQN was indeed the most reliable in the roundabout (highest baseline reward of 10.08) and showed moderate highway performance. Its improvements from tuning were consistently the smallest of the three algorithms, which is consistent with value-based methods having a lower adaptive ceiling in dynamic, multi-agent environments.

Conclusions

This project demonstrated three major findings about RL algorithm choice and hyperparameter tuning in autonomous driving environments.

First, algorithm choice matters, but in different ways depending on the environment. PPO is the most reliable algorithm for general use: its default configuration performs well, and it never degrades after tuning. A2C has the highest ceiling given correct tuning, but requires careful hyperparameter configuration to realize that potential. DQN provides consistent baseline performance but has a lower adaptive ceiling in complex environments.

Second, hyperparameter tuning is most impactful for algorithms with high sensitivity (A2C) and least impactful for algorithms that are already well-calibrated (PPO). The entropy coefficient was the single most impactful parameter across all environments: increasing it from 0.0 to 0.01–0.05 was responsible for the majority of meaningful behavioral improvements observed in the tuned experiments.

Third, environment difficulty creates a fundamental learning horizon that hyperparameter tuning cannot overcome without additional structural changes. The highway task is learnable within 25,000 timesteps by all three algorithms. The roundabout is partially learnable with a modest ceiling. The intersection is largely unsolvable within this budget. For the intersection, reward shaping (partial rewards for safe approach behavior, gap detection, or controlled deceleration) would be necessary to guide exploration toward the correct strategy.

These findings reflect the core theoretical framework covered in course the performance of any RL agent is a function of the algorithm, the hyperparameters, and the environment structure. There is no universally best algorithm only the best choice for a given problem structure, reward signal, and training budget.

General Reaction and Reflection

Project Reflection

This final project was one of the most difficult and time-consuming experiences of the course. The total training time exceeded 30 hours and that figure does not include time spent managing the numerous session disconnections that occurred throughout the project. Colab's tendency to reset RAM mid-run was the single most frustrating technical challenge. To handle this, safety nets were built throughout the code: before any training cell executes, it checks Google Drive for a saved checkpoint and resumes from the latest one if found. Without this, entire experiment sets would have needed to be rerun from scratch after each crash.

The decision to use 25,000 timesteps per run was a necessary compromise between training quality and total runtime. With 18 training runs (3 environments × 3 algorithms × baseline and tuned), and highway runs taking approximately 9 hours for all three algorithms, a longer budget would have been impractical. The 25,000-timestep constraint was sufficient to observe clear algorithmic differences on the highway and roundabout, but it was clearly insufficient for the intersection.

Despite the frustrations, the project was genuinely instructive in ways that reading about RL could not be. Building and running the full experimental pipeline training, checkpointing, evaluating, recording videos, generating charts gave a much deeper understanding of how RL agents actually behave compared to how they are described in theory. Watching the baseline videos was the most revealing experience: seeing all three agents drive straight into walls without any lane-switching behavior made the abstract concept of "exploration failure" suddenly very concrete and easy to diagnose. Watching the DQN tuned agent finally switch lanes in the highway felt like a genuine reward after 30 hours of work.

Course Reflection

This course provided a thorough and well-structured progression through reinforcement learning, from the theoretical foundations of multi-armed bandits and MDPs all the way to deep RL and RLHF. Each topic built directly on the previous one, and by the time the course reached PPO and actor-critic methods, the earlier theoretical work provided the exact context needed to understand why they were developed and why earlier methods fail in certain settings.

The homework assignments were instrumental in preparing for this project. The Q-learning snake game (HW2) provided hands-on experience with the exploration-exploitation tradeoff the same tension that dominated the analysis of the intersection environment in this project. The DQN Pong project (HW4) taught the practical realities of experience replay and target networks, and more importantly, the patience required for deep RL training. Managing a 30-hour training run without that prior experience would have been significantly harder. The RLHF assignment (HW5) connected the course to the current state of AI in a way that few other courses do.

The most important takeaway is that RL in practice is far messier than RL in theory. The gap between an algorithm's theoretical guarantees and its actual behavior on a real task is filled by the practical details: hyperparameter sensitivity, training budget constraints, sparse rewards, exploration failures, and the randomness of single-episode evaluation. This project transformed each of those concepts from abstract topics into concrete observations, and that grounding is the most valuable outcome of the course.